

Fixed-branching diamonds factor through the summing basis

A positive answer to Remark 1.7 of arXiv:1609.06618

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Status: candidate full solution, likely valid. For every fixed finite branching number k , we construct the factorization maps asked for in Remark 1.7. If C_2 is a binary-diamond constant, the construction gives $C(k) = 2C_2 \lceil \log_2 k \rceil$. The new ingredient is an elementary family of binary projections, one of which preserves each prescribed pairwise distance exactly.

1 Source question

Ostrovskii and Randrianantoanina study multibranching diamonds $D_{n,k}$ and factorization between the unit-vector and summing bases [1]. Their condition (1.1) asks for maps $f_n : D_{n,k} \rightarrow \ell_1$ such that, uniformly in n ,

$$\|f_n(u) - f_n(v)\|_1 \leq d_{D_{n,k}}(u, v) < C(k) \|f_n(u) - f_n(v)\|_s. \quad (1)$$

Here

$$\|(a_i)\|_s = \sup_r \left| \sum_{i=1}^r a_i \right|.$$

On page 4, Remark 1.7 asks whether $C(k)$ exists for every fixed k and emphasizes that even $C(3)$ was unknown:

assumption of Corollary 1.5 is equivalent (with modified constants) to the assumption: there exists an embedding $f : M \rightarrow \ell_1$ such that

$$\forall u, v \in M \quad \|f(u) - f(v)\|_1 \leq d_M(u, v) < C \cdot \|f(u) - f(v)\|_s. \quad (1.1)$$

We prove the following result (see Section 4).

Theorem 1.6. *For every $C > 1$ there exists $k(C) \in \mathbb{N}$ such that if for some $k \in \mathbb{N}$ and every $n \in \mathbb{N}$ there exists an embedding $f_n : D_{n,k} \rightarrow \ell_1$ satisfying*

$$\forall u, v \in D_{n,k} \quad \|f_n(u) - f_n(v)\|_1 \leq d_{D_{n,k}}(u, v) < C \cdot \|f_n(u) - f_n(v)\|_s,$$

then $k \leq k(C)$.

Remark 1.7. We note that Theorem 1.6 does not exclude the possibility that $\forall k \in \mathbb{N} \exists C = C(k) > 1$ so that for all $n \in \mathbb{N}$ there exists an embedding from $D_{n,k}$ into ℓ_1 that satisfies condition (1.1). Theorem 1.6 only implies that if such numbers $C(k)$ exist for all $k \in \mathbb{N}$ then they would not be uniformly bounded.

We do not know whether such numbers $C(k)$ exist for all $k \in \mathbb{N}$. Johnson and Schechtman [20] proved that $C(2)$ exists, but we don't even know whether $C(3)$ exists.

From another perspective, the results of the present paper can be viewed as a step in a generalization of results on existence of low-distortion embeddings of finite metric

We answer the entire fixed-branching question affirmatively.

2 Statement of the result

We use the recursive description of $D_{n,k}$. The graph $D_{0,k}$ is one edge with bottom and top endpoints. To obtain $D_{n,k}$, replace every edge by k paths of length two, with the edge lengths halved. Equivalently, $D_{n,k}$ is the parallel composition of k arms, each arm being the series composition of two copies of $D_{n-1,k}$ scaled by $1/2$.

Johnson and Schechtman proved the binary case [2]. Thus there is a constant $C_2 > 1$ such that, for every n , some $g_n : D_{n,2} \rightarrow \ell_1$ satisfies

$$\|g_n(a) - g_n(b)\|_1 \leq d_{D_{n,2}}(a, b) < C_2 \|g_n(a) - g_n(b)\|_s. \quad (2)$$

Theorem 1 (All fixed branchings). *For every integer $k \geq 2$ and every $n \in \mathbb{N}$, there is a map $F_{n,k} : D_{n,k} \rightarrow \ell_1$ satisfying*

$$\|F_{n,k}(x) - F_{n,k}(y)\|_1 \leq d_{D_{n,k}}(x, y) < 2C_2 \lceil \log_2 k \rceil \|F_{n,k}(x) - F_{n,k}(y)\|_s$$

for all distinct vertices x, y . In particular, the numbers $C(k)$ asked for in Remark 1.7 exist for every finite k , and one may take

$$C(k) = 2C_2 \lceil \log_2 k \rceil.$$

The logarithmic upper bound is compatible with Theorem 1.6 of [1], which implies that valid constants cannot remain bounded as $k \rightarrow \infty$. We make no claim that the logarithmic order or its coefficient is sharp.

3 Idea of the proof

Assign distinct binary words to the k arm labels. Reading one coordinate of those words collapses every k -fold parallel replacement onto a binary replacement. The resulting map to a binary diamond preserves the height of every vertex in every recursive cell. A pair of vertices has at most one relevant parallel split: the first recursive cell in which the two points occupy different arms. Any bit distinguishing those two arm labels preserves the pair's distance exactly.

We therefore have $m = \lceil \log_2 k \rceil$ nonexpansive binary images and at least one exact image for every pair. Compose all images with g_n , put them in consecutive coordinate blocks, and average by $1/m$. The ℓ_1 upper bound averages. An arbitrary offset from earlier blocks can reduce the summing norm visible inside the exact block by at most a factor two. This gives $2mC_2$.

4 Binary projections

Fix a function $q : \{1, \dots, k\} \rightarrow \{0, 1\}$. It induces recursively a map

$$\pi_q : D_{n,k} \longrightarrow D_{n,2}.$$

The bottom and top go to the binary bottom and top. In every replaced edge, the midpoint in source arm i goes to the midpoint in binary arm $q(i)$; inside each of the two child edges the same rule is repeated. Multiple source arms are allowed to collapse onto one binary arm.

Lemma 2 (One binary projection preserves each distance). *Every π_q is 1-Lipschitz and preserves the distance from a vertex to the bottom and top of every recursive subdiamond containing that vertex. Moreover, for every $x, y \in D_{n,k}$, one of the following holds:*

1. every map π_q preserves $d(x, y)$; or
2. there are two distinct arm labels $a, b \in \{1, \dots, k\}$ such that every q with $q(a) \neq q(b)$ satisfies

$$d_{D_{n,2}}(\pi_q(x), \pi_q(y)) = d_{D_{n,k}}(x, y).$$

Proof. We argue by induction on n , simultaneously for the endpoint assertion and the dichotomy. The result is immediate for $n = 0$. A final edge of $D_{n,k}$ maps to a final edge of $D_{n,2}$ of the same length, so π_q is 1-Lipschitz.

Let the bottom and top of the root diamond be B, T , and let M_i be the midpoint of arm i . Each lower piece $[B, M_i]$ and upper piece $[M_i, T]$ is a scaled copy of $D_{n-1,k}$. The induction hypothesis in these pieces shows that distance to their endpoints is preserved. Consequently the distance from any vertex to B and T is preserved as well. For example, if x is in the upper piece of arm i , the route from B through M_i has length

$$d(B, M_i) + d(M_i, x),$$

and no route entering through T is shorter; the same statement holds after projection. The other cases are symmetric. Applying the same reasoning inside any recursive cell proves the full endpoint assertion.

Now consider x, y . Cases in which one point is B or T are already covered by endpoint preservation. If x and y lie in distinct root arms $i \neq j$, put

$$h_x = d(B, x), \quad h_y = d(B, y), \quad L = d(B, T).$$

Different arms meet only at B and T , hence

$$d(x, y) = \min\{h_x + h_y, 2L - h_x - h_y\}. \quad (3)$$

If $q(i) \neq q(j)$, their images lie in distinct binary arms. Endpoint preservation gives the same h_x, h_y, L , so the binary analogue of (3) is identical. Alternative (2) holds with $a = i, b = j$.

It remains to suppose that both points lie in one root arm i . If they lie in the same lower piece or the same upper piece, apply the inductive dichotomy in that copy of $D_{n-1,k}$. If they lie in different pieces, say $x \in [B, M_i]$ and $y \in [M_i, T]$, every path either passes through M_i or exits through both B and T . The former route is no longer than the latter: writing $\alpha = d(B, x)$ and $\beta = d(y, T)$, their lengths are respectively

$$L - \alpha - \beta \quad \text{and} \quad L + \alpha + \beta.$$

Thus $d(x, y) = d(x, M_i) + d(M_i, y)$. Both terms are endpoint distances inside child diamonds and are preserved by every π_q . This is alternative (1) and completes the induction. $\square$

Choose pairwise distinct words

$$\sigma(1), \dots, \sigma(k) \in \{0, 1\}^m, \quad m = \lceil \log_2 k \rceil,$$

and let $q_j(i) = \sigma(i)_j$. Lemma 2 immediately gives the key metric identity

$$d_{D_{n,k}}(x, y) = \max_{1 \leq j \leq m} d_{D_{n,2}}(\pi_{q_j}(x), \pi_{q_j}(y)). \quad (4)$$

Indeed, in alternative (2), the distinct words of a and b differ in some coordinate; alternative (1) is even stronger. The reverse inequality in (4) follows from 1-Lipschitzness.

5 The summing norm of concatenated blocks

For finite sequences $z^{(1)}, \dots, z^{(m)}$, write $z = z^{(1)} \parallel \dots \parallel z^{(m)}$ for their concatenation.

Lemma 3 (Block visibility). *For every block j ,*

$$\|z\|_s \geq \frac{1}{2} \|z^{(j)}\|_s.$$

Proof. Let A be the sum of all coordinates preceding block j , and let P_r be the partial sums within that block, with $P_0 = 0$. Choose r with $|P_r| = \|z^{(j)}\|_s =: s$. The global partial sums include both A and $A + P_r$. Since

$$s = |P_r| \leq |A| + |A + P_r|,$$

one of those two global partial sums has absolute value at least $s/2$. $\square$

6 Proof of Theorem 1

For $x \in D_{n,k}$ define

$$F_{n,k}(x) = \frac{1}{m} (g_n(\pi_{q_1}(x)) \parallel \dots \parallel g_n(\pi_{q_m}(x))). \quad (5)$$

Because g_n and every π_{q_j} are nonexpansive in the relevant ℓ_1 and graph metrics,

$$\begin{aligned}\|F_{n,k}(x) - F_{n,k}(y)\|_1 &= \frac{1}{m} \sum_{j=1}^m \|g_n(\pi_{q_j}(x)) - g_n(\pi_{q_j}(y))\|_1 \\ &\leq \frac{1}{m} \sum_{j=1}^m d(\pi_{q_j}(x), \pi_{q_j}(y)) \leq d(x, y).\end{aligned}$$

By (4), choose j for which $d(\pi_{q_j}(x), \pi_{q_j}(y)) = d(x, y)$. Lemma 3 and the binary estimate (2) give

$$\begin{aligned}\|F_{n,k}(x) - F_{n,k}(y)\|_s &\geq \frac{1}{2m} \|g_n(\pi_{q_j}(x)) - g_n(\pi_{q_j}(y))\|_s \\ &> \frac{d(x, y)}{2mC_2}.\end{aligned}$$

This is the claimed strict inverse inequality and completes the proof.

7 Verification, scope, and novelty

The packet includes an independent address-based exhaustive checker. It constructs all bit projections, computes all-pairs shortest-path distances, and verifies (4) at finite depths before checking the concatenated Hamming realization. This is only a regression test; the inductive proof above covers all n .

The result answers exactly the full fixed-branching question in Remark 1.7, including the highlighted case $k = 3$. It does not solve the neighboring Planar Conjecture, forbidden-minor conjecture, or the problem of embedding all series-parallel metrics into every non-superreflexive Banach space.

Before promotion, the run’s solution, attempt, proof-gap and registry indexes were searched by arXiv id, exact question, and the phrases “multibranching diamond”, “summing norm”, “ $D_{n,3}$ ”, and “ $C(3)$ ”. A bounded primary arXiv search used the same phrases together with “binary projection” and “fixed branching”. It found no later resolution or this binary-code construction. This is not an exhaustive novelty certification; the separate `novelty.md` records the search bounds.

Human review is strongly recommended. The principal audit target is Lemma 2, especially the same-arm/different-half case. The new proof otherwise uses only the published binary case and elementary norm estimates.

References

- [1] M. I. Ostrovskii and B. Randrianantoanina, *A new approach to low-distortion embeddings of finite metric spaces into non-superreflexive Banach spaces*, arXiv:1609.06618 (2016), Remark 1.7.
- [2] W. B. Johnson and G. Schechtman, *Diamond graphs and super-reflexivity*, J. Topol. Anal. **1** (2009), 177–189, doi:10.1142/S1793525309000114.